Exam 2 Review

Ch29	The Monetary System
Ch30	Money Growth and Inflation
Ch33	Aggregate Demand and Aggregate Supply
Ch28	Unemployment

CHAPTER 29

The Monetary System

Multiple Choice

7 Bank T-Account
13 Currency Holding · Reserves
14 Excess Reserves · Loans
15 Bank Balance Sheet · Loans

Short Answer

5 Money Multiplier
6 Leverage · Bank Capital
10 Money Multiplier · Reserve Requirement
11 Open-Market Operations

14. People hold $400 million of bank deposits but no currency. Banks have made $380 million dollars of loans and only hold enough reserves to satisfy reserve requirements. Because of uncertainty, **banks choose to hold $10 million more in reserves.** The Fed takes no action. What happens to bank loans?

a. they fall $220 million
b. they fall $200 million
c. they rise $200 million
d. they rise $220 million

Money Multiplier Stays;
$r = 20/400 = 0.05(5\%)$
$1/r = 20$

Deposit changes by 10:
$-\$10 \times 20 = -\$200$

10. Assume that the banking system has total reserves of $100 billion. Assume also that **required reserves are 10 percent** of checking deposits and that banks hold no excess reserves and households hold no currency.

a. What is the money multiplier? What is the money supply?

b. If the **Fed now raises required reserves to 20 percent of deposits**, what are the change in reserves and the change in the money supply?

Money Multiplier Changes

$1/0.1 = 10 \rightarrow 1/0.2 = 5$

M1: Currency(지폐), Demand deposit, etc
M2: M1 + Saving deposits, etc

5. You take **$100 you had kept under your mattress** and deposit it in your bank account. If this $100 stays in the banking system as reserves and if banks hold reserves equal to 10 percent of deposits, by how much does the total amount of deposits in the banking system increase? **By how much does the money supply increase?**

Money Supply
= Currency + Demand Deposits

시장의 통화량에 영향을 주는 것:
Monetary Policy(reserve requirements, OMO)

$100 already in the money supply.
Deposit + 100 : $100 × 1/10 = $1,000

Money supply increase by $900
(+$1,000 if $100 is 본원통화)

11. Assume that the reserve requirement is 20 percent. Also, assume that banks do not hold excess reserves and there is no cash held by the public. The Fed decides that it wants to expand the money supply by $40 million.

a. If the Fed is using **open-market operations**, will it buy or sell bonds?

Open-Market Operation:
본원통화(M_0) 조절 수단

b. What quantity of bonds does the Fed need to buy or sell to accomplish the goal? Explain your reasoning.

$1/0.2 \times$ 본원통화 = \$40

Scenario 29-1

The monetary policy of Namdian is determined by the Namdian Central Bank. The local currency is the dia. Namdian banks collectively hold 100 million dias of required reserves, 25 million dias of excess reserves, 250 million dias of Namdian Treasury Bonds, and their customers hold 1,000 million dias of deposits. Namdians prefer to use only demand deposits and so the money supply consists of demand deposits.

Assets	Liabilities
Required reserves Excess reserves	Deposits
Loans	
Treasury bonds	Bank Capital

15. **Refer to Scenario 29-1.** Assuming the only other item Namdian banks have on their balance sheets is loans, what is the value of existing loans made by Namdian banks?

a. 625 million dias
b. 875 million dias
c. 1,125 million dias
d. None of the above is correct.

6. Happy bank starts with $200 in bank capital. It then accepts $800 in deposits. It keeps 12.5 percent (1/8th) of deposits in reserve. It uses the rest of its assets to make bank loans.

a. Show the balance sheet of Happy bank. **Asset / Capital**
b. What is Happy Bank's leverage ratio? =1000/200 = 5

c. Suppose that 10 percent of the borrowers from Happy Bank default and these bank loans become worthless. Show the bank's new balance sheet.
d. By what percentage do the bank's total assets decline? By what percentage does the bank's capital decline? Which change is larger? Why?

Assets		Liabilities	
Reserves	100	Deposits	800
Loans	900		
Treasury bonds		Bank Capital	200
↓		↓	
Loans	810	Bank Capital	110

9%, 45%. **Defaulted loans are covered by bank capital.**
(= 9% × Leverage Ratio)

CHAPTER 30

Money Growth and Inflation

Multiple Choice

- 10 Classical Dichotomy
- 21 Unexpected Inflation · Redistribution
- 8 Velocity of Money
- 9 Value of Money · Inflation Rate
- 19 Inflation · Capital Gains Tax

Short Answer

- 1 Quantity Theory · Money Supply Rule
- 2 Money Demand Shift · Price Level
- 4 Relative Prices · Inflation
- 5 After-Tax Real Interest Rate

10. According to the **classical dichotomy**, which of the following is not influenced by monetary factors?

a. nominal GDP and nominal interest rates
b. **real** wages and **real** GDP
c. the price level and nominal GDP
d. None of the above is correct.

통화 ↛ 실질

21. Wealth is redistributed from debtors to creditors when inflation is

a. high, whether it is expected or not.
b. low, whether it is expected or not.
c. unexpectedly high.
d. unexpectedly low.

Expected inflation → interest rates에 반영

Debtor(채무자): 예상보다 높으면 이득 (덜 냄)
Creditor(채권자): 예상보다 낮으면 이득 (채무자가 더 냄)

Figure 30-2. On the graph, MS represents the money supply and MD represents money demand. The usual quantities are measured along the axes.

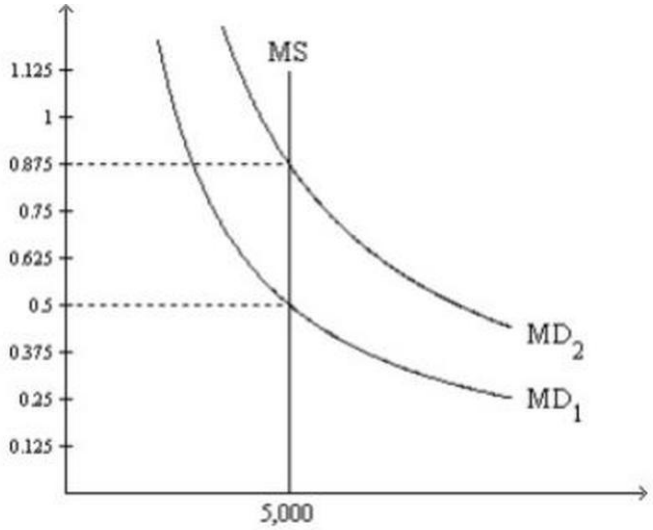
8. **Refer to Figure 30-2.**
Suppose the relevant money-demand curve is the one labeled MD_1; also suppose the economy's real GDP is 30,000 for the year.
If the money market is in equilibrium, then how many times per year is the typical dollar bill used to pay for a newly produced good or service?

a. 4 b. 6 c. 8 d. 12

9. **Refer to Figure 30-2.**
At the end of 2009 the relevant money-demand curve was the one labeled MD_2.
At the end of 2010 the relevant money-demand curve was the one labeled MD_1.
Assuming the economy is always in equilibrium, what was the economy's approximate inflation rate for 2010?

a. -43 percent b. -57 percent c. 57 percent d. 75 percent

1/P



$P_1 = 1/0.875$

$P_2 = 1/0.5$

Inflation: $(P_2 / P_1) - 1$

5. if the tax rate is 40 percent, compute the before-tax real interest rate and the **after-tax real interest rate** in each of the following cases.

a. The nominal interest rate is 10 percent, and the inflation rate is 5 percent.

b. The nominal interest rate is 6 percent, and the inflation rate is 2 percent.

c. The nominal interest rate is 4 percent, and the inflation rate is 1 percent.

Taxed on the nominal rate

$10\% \times (1 - 0.4) - 5\%$

19. The country of Robinya has a tax system identical to that of the United States. Suppose someone in Robinya bought a parcel of land for 10,000 deera (the local currency) in 1970 when the price index equaled 100. In 2010, the person sold the land for 100,000 deera, and the price index equaled 500. The tax rate on nominal capital gains was 20 percent. Compute the taxes the person paid on the nominal gain and the change in the real value of the land in terms of 2010 prices to find the after-tax real rate of capital gain.

a. -20 percent b. 20 percent c. 42 percent d. 64 percent

(1) Tax on Nominal Gain:
(100,000 – 10,000) × 0.2 =18,000

(2) 1970 price to 2010 price
10,000 × 500/100 = 50,000

(3) Change in the Real Value
(100,000 – 50,000 – 18,000) / 50,000 = 0.64

해설 생략

1. The money supply is $500 billion, nominal GDP is $10 trillion, and real GDP is $5 trillion.

$$M V = P Y$$

가격수준 × 실질생산량
= 명목GDP

a. Find the price level and the velocity of money.

b. If velocity is constant and real output rises 5% a year, what happens to nominal GDP and the price level next year if the Fed keeps the money supply constant?

$$M V = P Y$$

-5% 5%

Nominal GDP : +0%

c. What money supply keeps the price level stable next year?

+5%

d. What money supply gives 10% inflation next year?

+15%

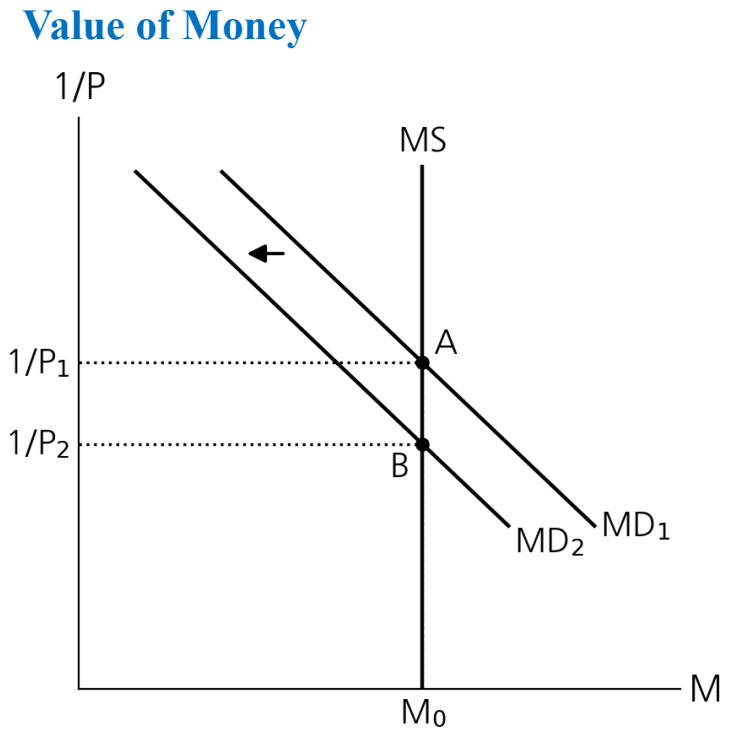
10% 5%

2. Suppose that changes in bank regulations **expand the availability of credit cards** so that people need to hold less cash.

a. How does this affect the demand for money?

b. If the Fed does not respond to this event, what will happen to the price level?

c. If the Fed wants to keep the price level stable, what should it do?



CHAPTER 33

Aggregate Demand and Aggregate Supply

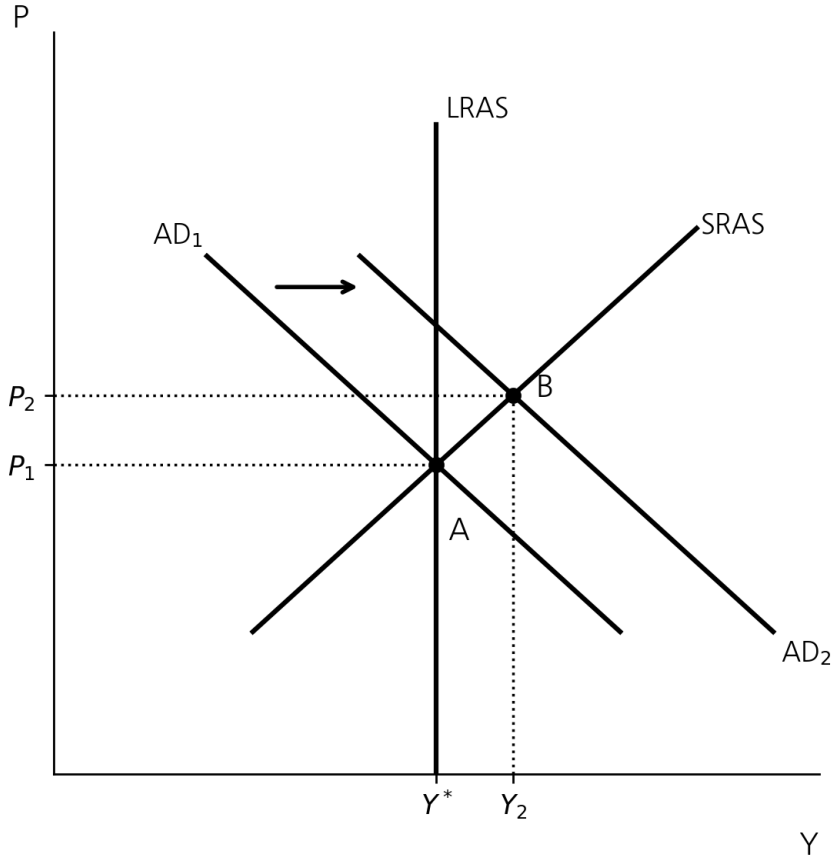
Multiple Choice

11 Expected Price Level · SRAS
16 Multiple Shocks · AD Shifts
7 Capital Inflow · Exchange Rate
8 Net Exports · AD
17 Stock Market Boom · Long-Run Equilibrium

Short Answer

1 Long-Run Aggregate Supply Shifters
2 Money Neutrality · Sticky Wages
4 Short-Run vs. Long-Run Effects
5 Investment Boom · AD–AS

Ch33 AD–AS Overview · Curve Shifters



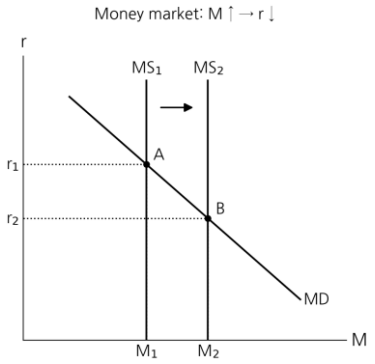
Change in P = movement along AD / SRAS, not a shift

Why AD Slopes Downward: P에 의한 변화

- Wealth Effect (P↓ → real wealth↑ → C↑, Y↑)
- Interest-Rate Effect (P↓ → MD↓ → r↓ → I↑, Y↑)
- Exchange-Rate Effect (P↓ → r↓ → capital outflow → depreciation → NX↑, Y↑)

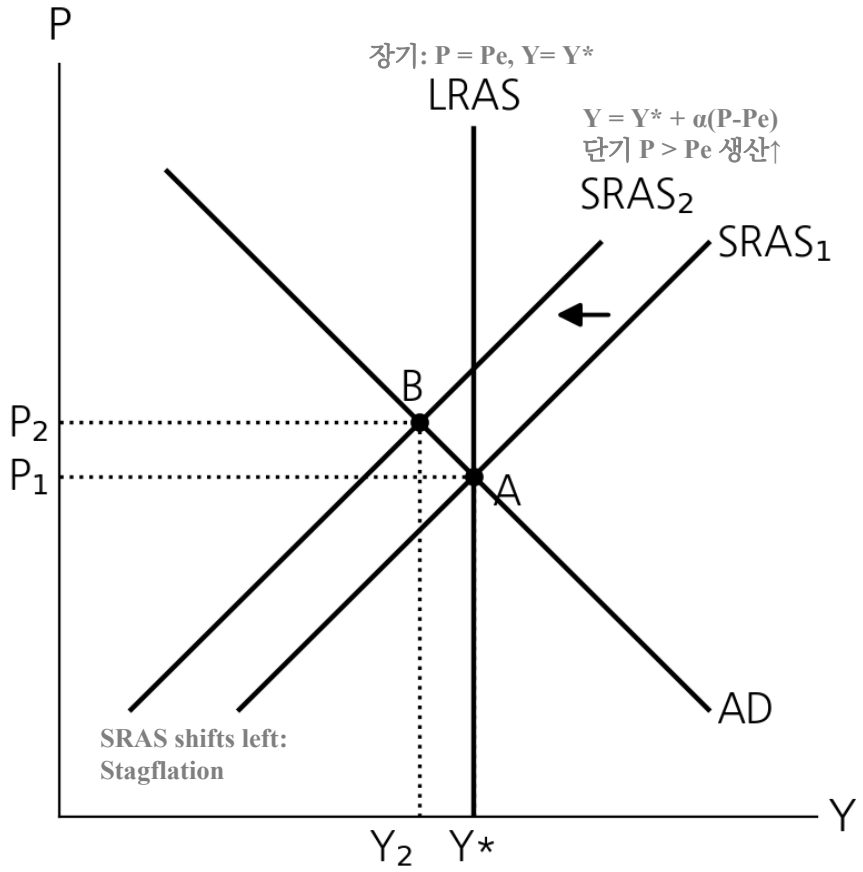
AD Shifters: C, I, G, NX changes

- C: Stock boom → wealth ↑, income taxes, consumer expectations
- I: M ↑ → r ↓, investment tax credit, firm expectations
- G: Government purchases
- NX: Exchange rate, foreign income



Ch33 AD–AS Overview · Curve Shifters

Y*: Natural rate of output,
full-employment output, potential output



Change in P = movement along AD / SRAS, not a shift

Why SRAS Slopes Upward: P에 의한 변화

- Sticky-Wage (**P↑ → real wage↓** → hire↑, Y↑)
- Sticky-Price (**P↑ → relative price of sticky firms↓** → sales↑, Y↑)
- Misperception (**P↑ → firms see higher relative price** → produce↑, Y↑)

SRAS Shifters: Pe, Cost, LRAS factors

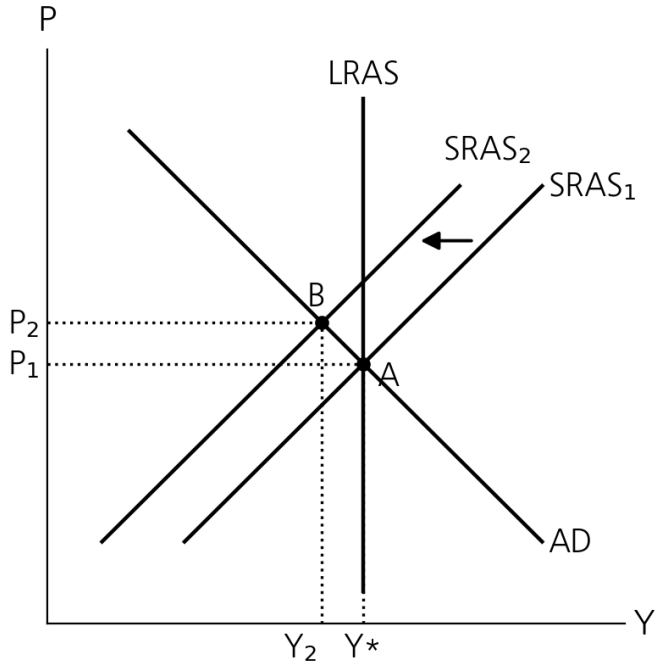
- Expected price level ↑ → [SRAS ←]
- Nominal wages, input costs(ex. oil shock) ↑ → [SRAS ←]
- Productivity ↑→ [LRAS & SRAS →]

LRAS Shifters

- L, K, H, N, technology ↑ → [LRAS →]
- Y* ↑ (natural output)
- M or P alone: no shift

11. An increase in the **expected price level** shifts the

a. short-run and long-run aggregate supply curves left.
b. the short-run but not the long-run aggregate supply curve left.
c. the long-run but not the short-run aggregate supply curve left.
d. neither the long-run nor the short-run aggregate supply curve left.



Political Instability Abroad **Factor other than P → AD Shifter** (vs P's Exchange-Rate Effect → AD Slopes Downward)

Suppose that political instability in other countries makes people fear for the value of their assets in these countries so that they desire to purchase more Korean assets.

7. **Refer to Political Instability Abroad.** What would happen to the Korean won?
a. It would **appreciate in foreign exchange markets** making Korean goods more expensive compared to foreign goods.

What would the change in the exchange rate make happen to **Korean net exports and Korean aggregate demand?**
d. Net exports would fall which by itself would decrease Korean aggregate demand.

평가절상 원화가치 상승 달러가치 하락 → 수입 증가 수출 감소 → Y=C+I+G+**NX↓** → **AD 하락(왼쪽)**

1. Explain whether each of the following events will increase, decrease, or have no effect on **long-run aggregate supply.** **L, N, H, K**

a. The United States experiences a **wave of immigration.** **L**

b. Congress raises the **minimum wage** to $15 per hour. **Natural Rate of Unemployment → LRAS**

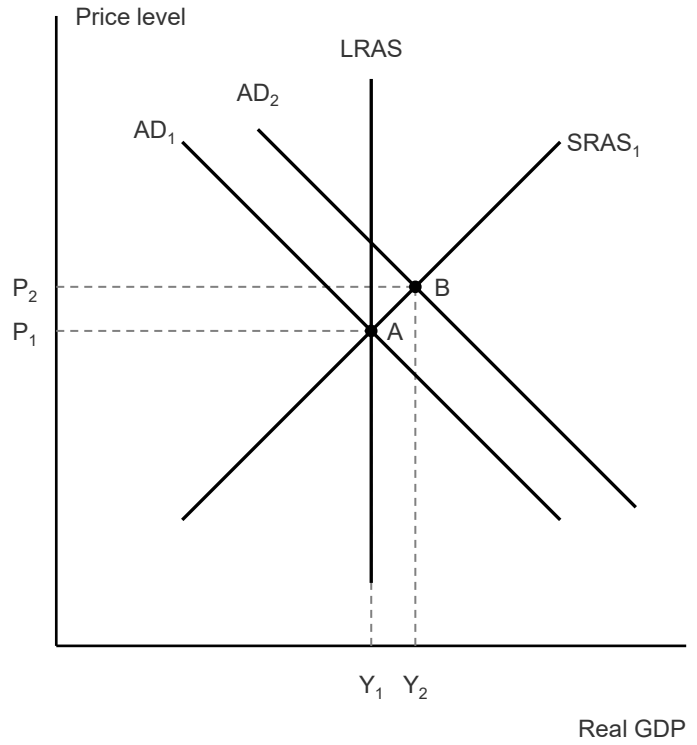
c. Intel invents a new and more **powerful computer chip.** **Productivity**

d. A severe **hurricane damages** factories along the East Coast. **Capital Stock(K)**

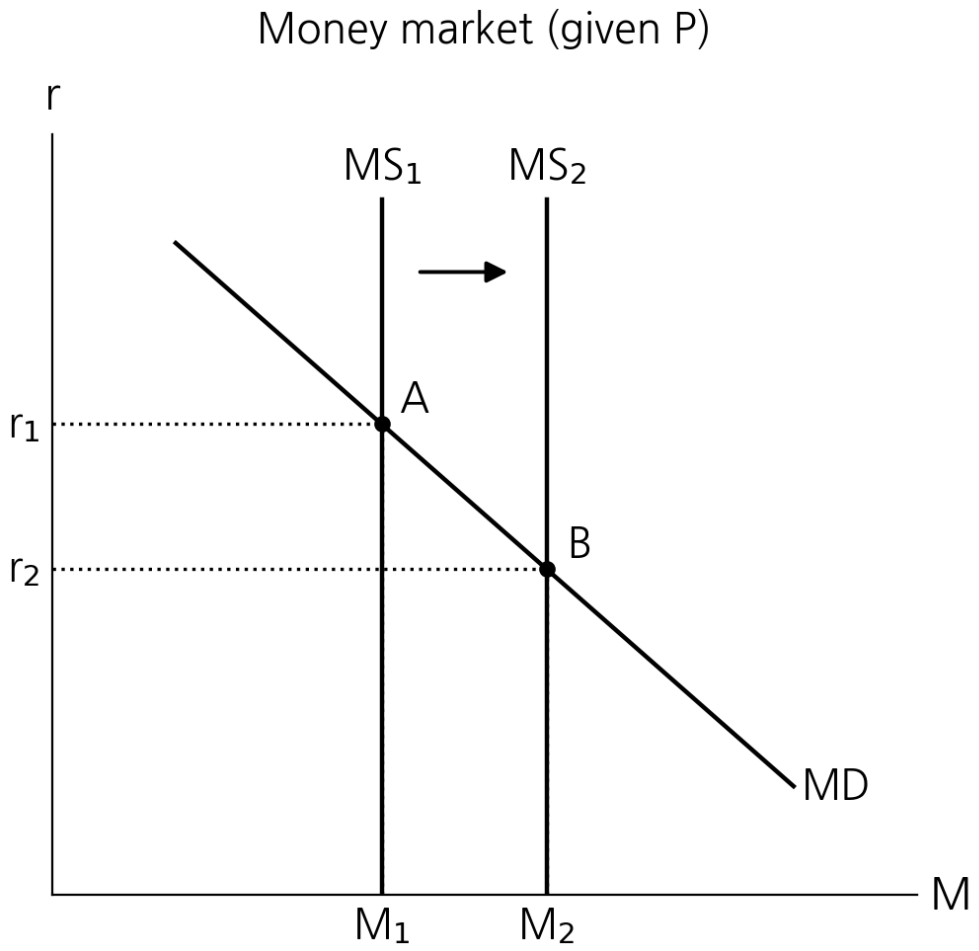
2. Suppose an economy is in long-run equilibrium.

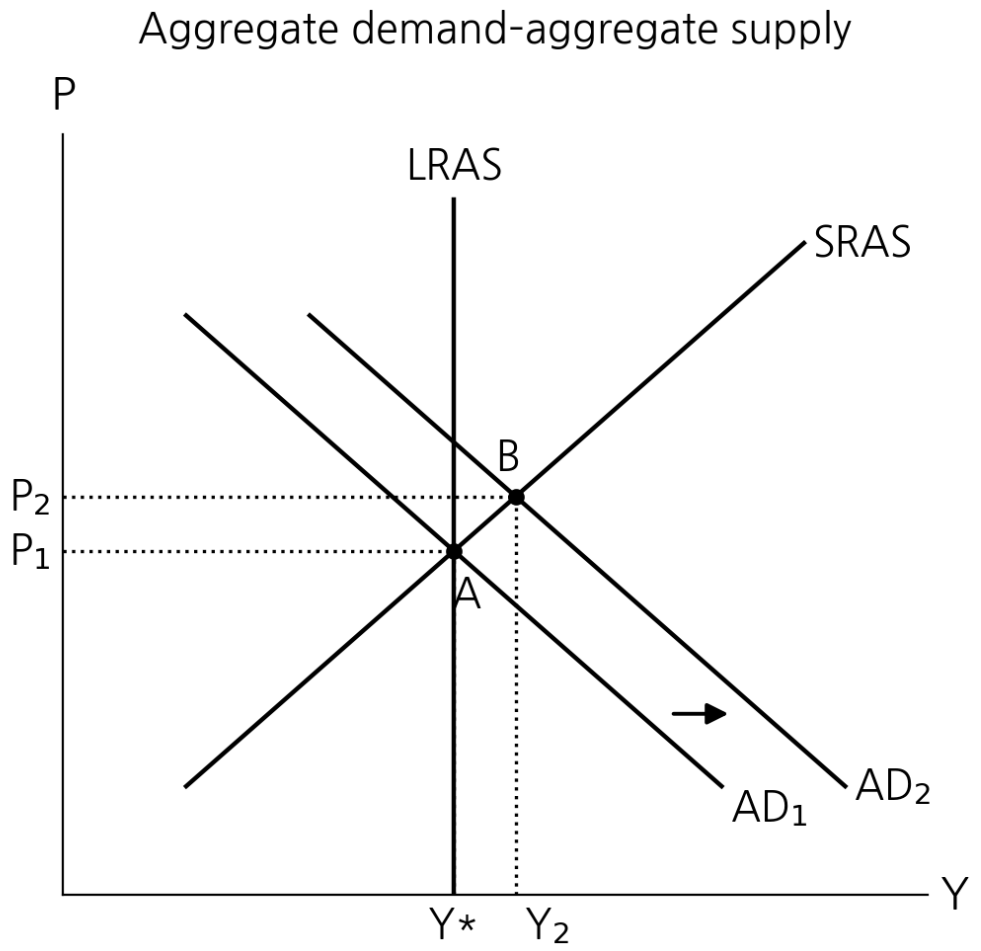
a. Use the model of aggregate demand and aggregate supply to illustrate the initial equilibrium (call it point A). Be sure to include both short-run and long-run aggregate supply.

b. **The central bank raises the money supply by 5** percent. Use your diagram to show what happens to output and the price level as the economy moves from the initial to the new short-run equilibrium (call it point B).



Ch33 Money Supply → Aggregate Demand





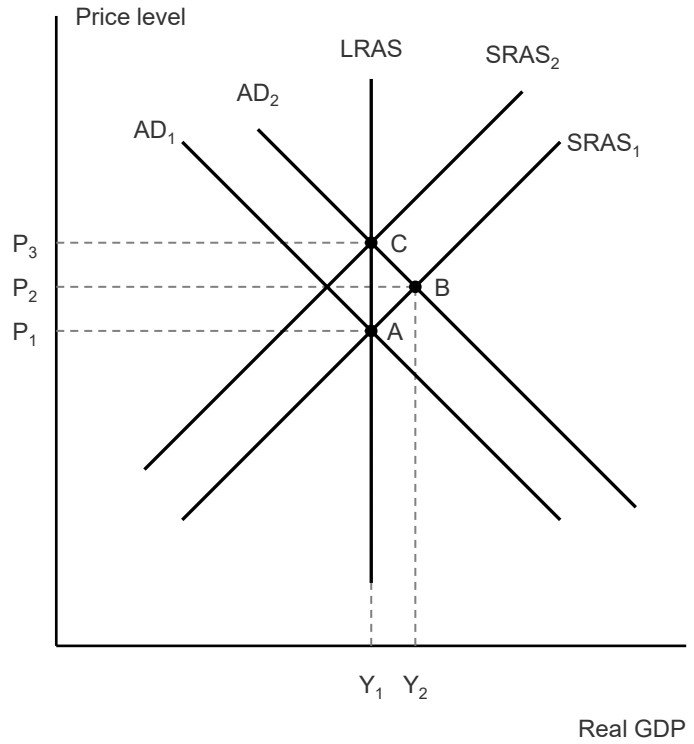
MS ↑ → r ↓ → Investment ↑

→ AD shifts right → Short-run Y ↑ and P ↑

2. Suppose an economy is in long-run equilibrium.

c. Now show the new long-run equilibrium (call it point C). What causes the economy to move from point B to point C?

d. According to the sticky-wage theory of aggregate supply, how do nominal wages at point A compare to nominal wages at point B? How do nominal wages at point C?

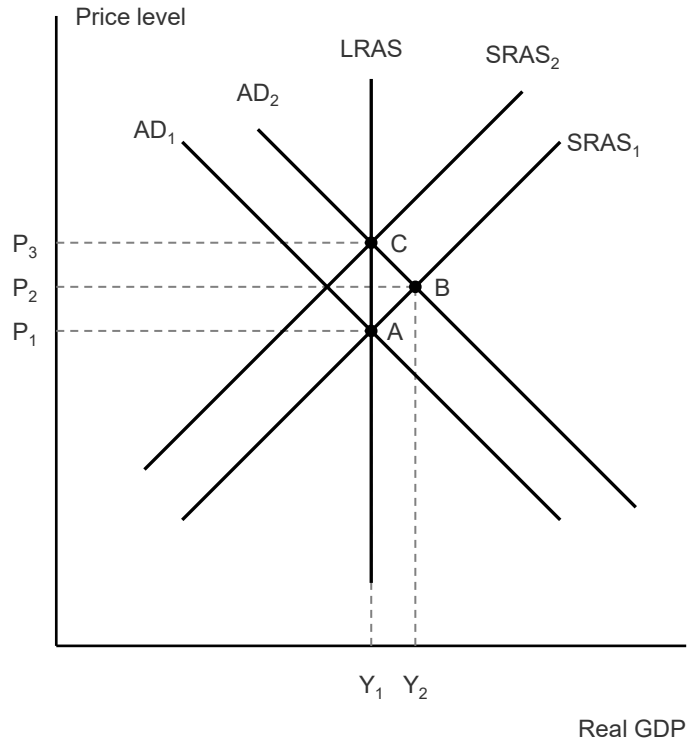


Nominal Wage **sticky in the short run** (same at A, B)
Nominal Wage **adjust to higher price level** (higher at C)
As production cost increase, LRAS shift left.

2. Suppose an economy is in long-run equilibrium.

e. According to the sticky-wage theory of aggregate supply, how do real wages at point A compare to real wages at point B? How do real wages at point A compare to real wages at point C?

f. Judging by the impact of the money supply on nominal and real wages, is this analysis consistent with the proposition that money has real effects in the short run but is neutral in the long run?



At B, real wage decrease as nominal wage stays the same and P increase
At C, real wage is back to A as nominal wage adjusts.

5. Suppose firms become very optimistic about future business conditions and **invest heavily in new capital equipment.**

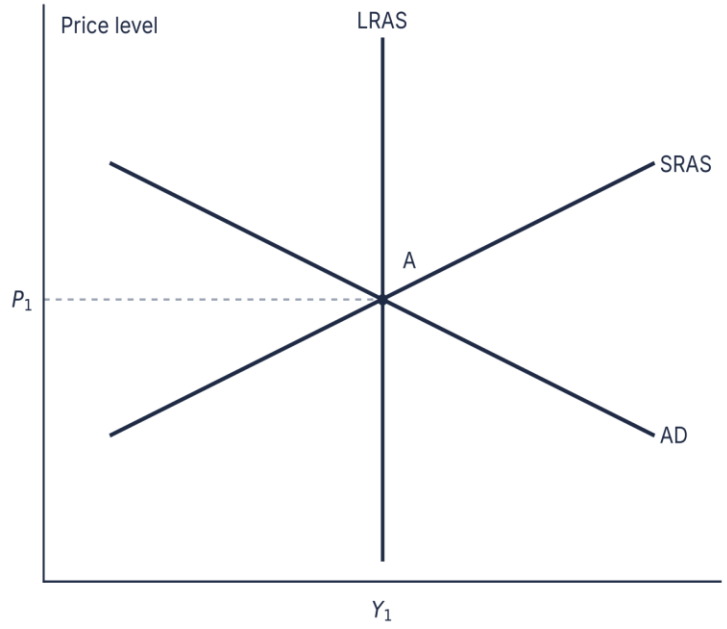
a. Draw an AD–AS diagram to show the short-run effect of this optimism. Label the new price level and real output. Explain in words why the **aggregate quantity of output supplied** changes.

= why does SRAS slopes upward?

b. Using the diagram from part (a), show the new long-run equilibrium (assume no change in LRAS). Explain in words why the aggregate quantity of output demanded changes between the short run and the long run. **SRAS moves to the left, as wage, price, misperception adjusts**

c. How might the investment boom affect the long-run aggregate-supply curve? Explain.

Might shift right as Capital Stock increases.



4. **(Short Answer 4) Practice**

a. The stock market declines sharply, reducing consumers' wealth.

b. The federal government increases spending on national defense. (vs tax cut)

c. A technological improvement raises productivity.

d. A recession overseas causes foreigners to buy fewer U.S. goods.

CHAPTER 28

Unemployment

Multiple Choice

1 Labor-Force Participation
5 Classification · Unemployment Rate
10 Unions · Wages and Employment
14 Efficiency Wages · Natural Rate

Short Answer

1 Labor Force Statistics
2 Employment vs. Unemployment
4 Sectoral Shifts · Wage Adjustment

1. Sheamous loses his job and decides to sit on the beach rather than look for work during the next few months. Other things the same, the unemployment rate

a. increases and the labor-force participation rate decreases.
b. increases and the labor-force participation rate is unaffected.
c. is unaffected and the labor-force participation rate decreases.
d. and the labor-force participation rate are both unaffected.

$$\frac{\text{Unemployed}}{\textbf{Employed} + \text{Unemployed}}$$

$$\frac{\textbf{Labor Force}}{\text{Adult population}}$$

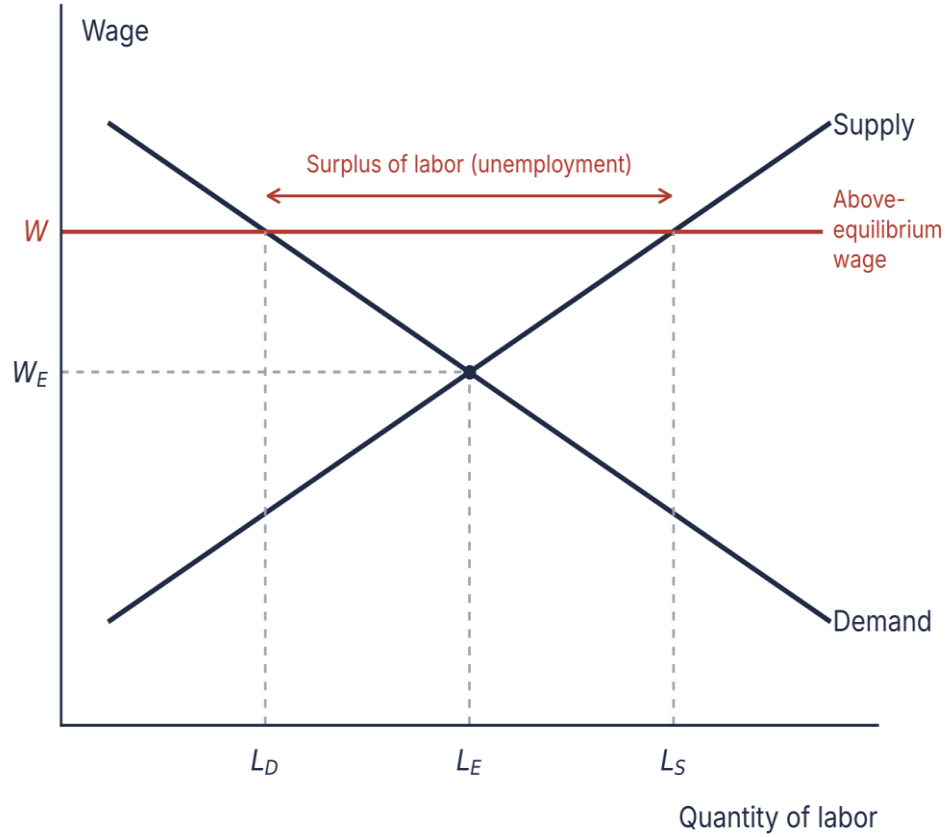
5. Suppose that some people are counted as unemployed when, to maintain unemployment compensation, they search for work only at places where they are unlikely to be hired. If these individuals were counted as out of the labor force instead of as unemployed, then

a. both the unemployment rate and labor-force participation rate would be higher.
b. both the unemployment rate and labor-force participation rate would be lower.
c. the unemployment rate would be lower and the labor-force participation rate would be higher.
d. the unemployment rate would be higher and the participation rate would be lower.

$$\frac{\textbf{Unemployed}}{\text{Employed} + \textbf{Unemployed}}$$

10. Suppose that autoworkers are unionized in one region of the country but not in the other. If the **(1)unionized workers** abolished their unions, employment

a. and wages of autoworkers in that region would rise.
b. of automobile workers in that region would rise, but wages of automobile workers in that region would fall.
c. and wages of autoworkers in that region would fall.
d. of automobile workers in that region would fall but wages of automobile workers in that region would rise.



+ **(2)Efficiency wages contribute to**
c. structural unemployment and the **natural rate** of unemployment.

natural rate of unemployment = structural + frictional (cyclical X)

+ What if the **(3)minimum wage is set below** the equilibrium wage?

1. In June 2009, at the trough of the Great Recession, the Bureau of Labor Statistics announced that of all adult Americans, 140,196,000 were employed, 14,729,000 were unemployed, and 80,729,000 were not in the labor force. Use this information to calculate:

a. the adult population

b. the labor force

c. the labor-force participation rate

d. the unemployment rate

Adult population		
Not in..	Labor force	
	Unemployed	Employed

+ (MC 4) What is the adult female labor-force participation rate?

전체(분모)를 adult female 만 대상으로
(여성 인구 중 참가율)

4. Frictional unemployment is said to result from a mismatch between employers' and workers' skills. Consider two industries: auto and aircraft manufacturing.

a. If workers can choose which industry to train for (similar training needed), what happens to wages in the two industries? How long does it take?

임금이 동일해질 때까지 노동자가 다른 산업으로 이동 (낮은임금 → 높은임금)

b. If the economy starts importing autos and exporting aircraft, what happens to labor demand in each industry?

수입 → 국내산업 쇠퇴 → 노동수요 감소

c. If workers cannot be quickly retrained, how do these shifts affect equilibrium wages in the short run and the long run?

Workers cannot move between industries;
Wage differs in SR(higher in aircraft),
equals in LR after training.

d. If wages fail to adjust to the new equilibrium, what occurs?

균형이 아닐 경우 임금 높은 곳에
노동의 초과공급 (=실업)

Answers 1 / 2 Ch29 · Ch30

Ch29

Multiple Choice

14 b
15 a

Short Answer 5

Deposits +$1,000
Money supply +$900

Short Answer 6

a. Reserves $100, Loans $900
Deposits $800, Capital $200
b. Leverage = 5
c. Loans $810, Capital $110
d. Assets −9%, Capital −45%

Short Answer 10

a. 10 · $1,000B
b. ΔR = 0, ΔM = −$500B

Short Answer 11

a. Buy bonds
b. $8 million

Ch30

Multiple Choice

10 **b**
21 **d**
8 **d**
9 **d**
19 **d**

Short Answer 1

a. P = 2, V = 20
b. P −4.76%; nominal GDP same
c. M +5% ($525B)
d. M +15.5% ($577.5B)

Short Answer 2

a. MD shifts left
b. P rises
c. Cut MS

Short Answer 5

Real: pre-tax → after-tax
a. 5.0% → 1.0%
b. 4.0% → 1.6%

c. 3.0% → 1.4%

Answers 2 / 2 Ch33 · Ch28

Ch33

Multiple Choice

11	b
7	a
8	d

Short Answer 1

a. ↑ b. ↓ c. ↑ d. ↓

Short Answer 2

b. Y↑, P↑
c. Y back, P↑ (SRAS ←)
d. Nominal A = B < C
e. Real B < A = C
f. Yes

Short Answer 4

a. SR Y↓P↓ · LR Y same, P↓
b. SR Y↑P↑ · LR Y same, P↑
c. Y↑, P↓
d. SR Y↓P↓ · LR Y same, P↓

Short Answer 5

a. AD →: Y↑, P↑
b. SRAS ←: Y back, P↑
c. LRAS may shift →

Ch28

Multiple Choice

1	a
5	b
10	b
14	c

10 +

Below W*: non-binding, no surplus
* UR ↓; LFPR ↓

Short Answer 1

a. 235,654,000
b. 154,925,000
c. 65.7%
d. 9.5%
+ (MC 4) Female LF ÷ female pop.

Short Answer 4

a. Wages equalize
b. D auto ↓, aircraft ↑
c. SR: auto ↓, aircraft ↑
LR: wages equalize
d. Aircraft shortage, auto surplus